

From a physics model to a computational experiment

Week 1 | Programming fundamentals through radioactive decay



A sample has a half-life of 6.0 h.

How should we simulate one day — and decide whether the answer deserves trust?



model



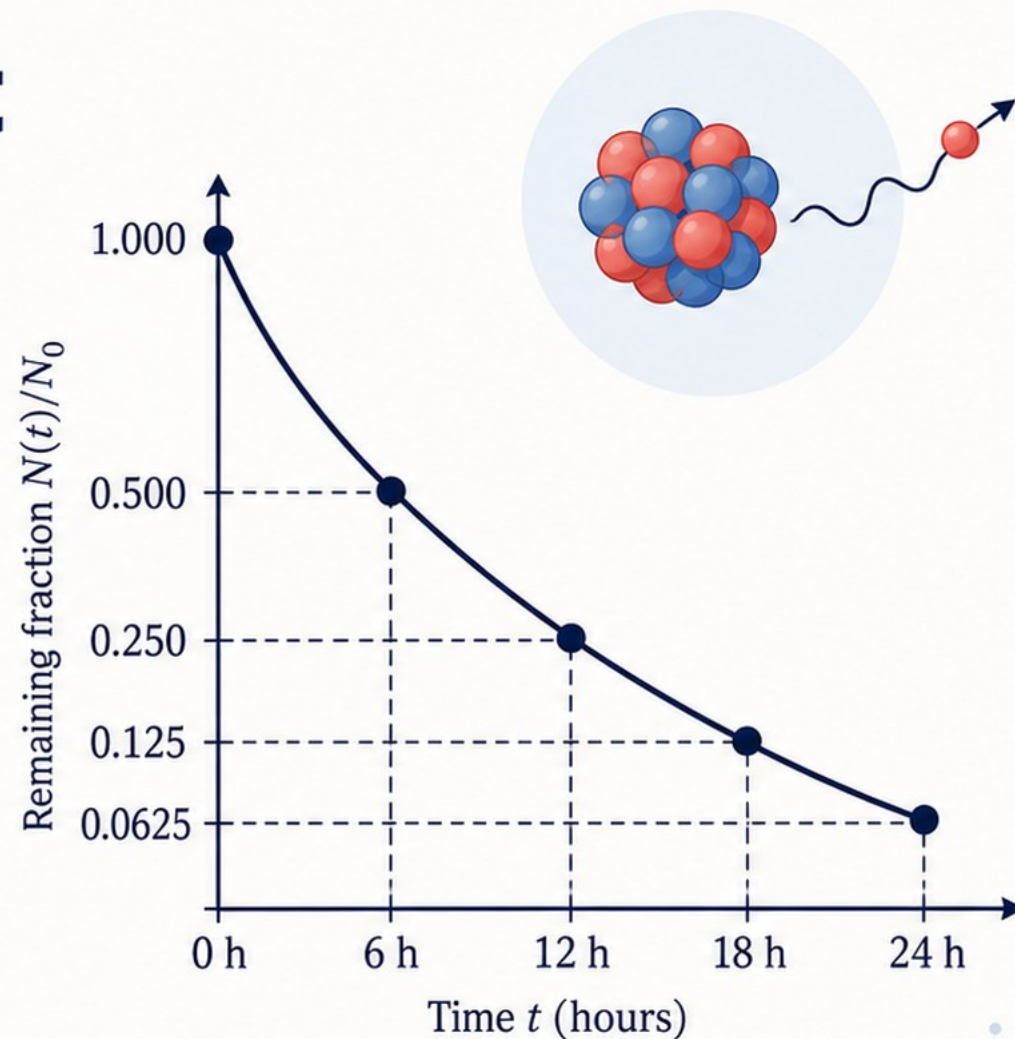
units



algorithm

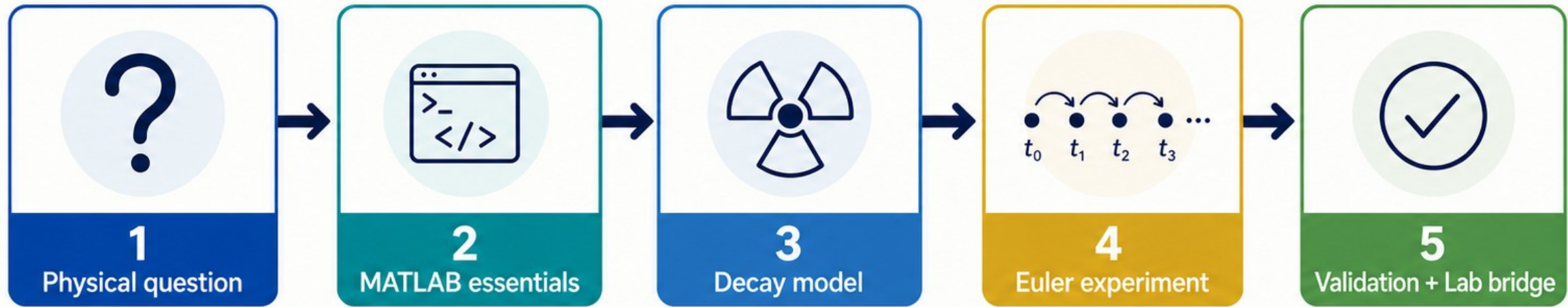


evidence



Today's arc: ask → compute → test

A physics question becomes a numerical experiment



Learning outcomes

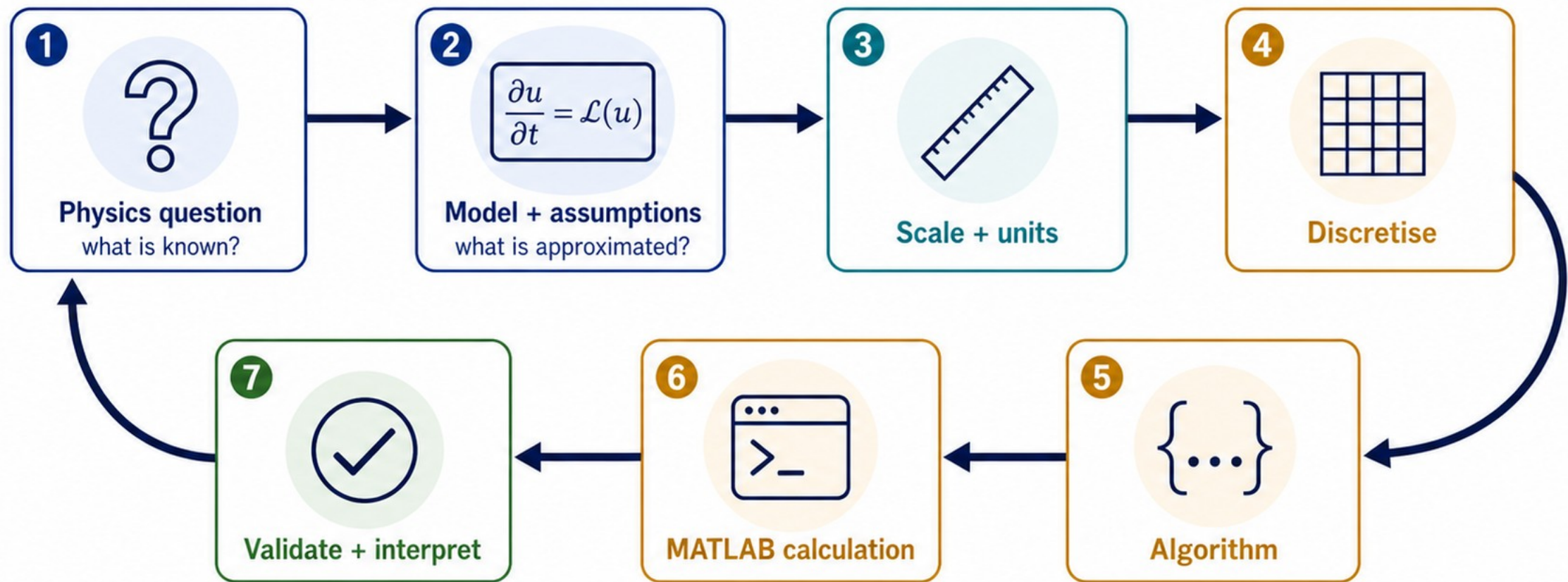
- ✓ Map physical quantities to variables and units
- ✓ Implement one readable time-step loop
- ✓ Compare with a reference and justify dt
- ✓ Record evidence so another person can rerun it



Course roadmap: Week 1 foundations → later numerical methods

Computation is a controlled experiment on a model

Every arrow carries a decision that should remain visible.



A graph alone is not evidence.

Four MATLAB views, one repeatable experiment

Know where to try, save, inspect, and rerun your work.



Command Window

Try one expression and inspect its answer.



Editor / Live Editor

Save the sequence of choices: text + code + output.



Workspace

Inspect variables currently in memory — useful, but easy to treat as hidden input.



Current Folder

Keep the script and related files together.



Use the Editor / Live Editor for work you intend to rerun.

Readable names make units and assumptions visible

The syntax is small; the naming discipline is the habit.

```
N0 = 1000;           % initial population, nuclei
```

```
T_half_h = 6.0;      % half-life, h
```

```
lambda_per_h = log(2)/T_half_h; % decay constant, 1/h
```



Assignment

Use = to store a value in a named variable.



Operators

Use arithmetic to translate an expression.



Comments

Use % to explain a choice to the next reader.



Unit naming

A suffix such as _h or _m2 keeps dimensions visible.



Unit check: $\text{lambda_per_h} \times \text{dt_h}$ must be dimensionless.

Onramp skills become evidence in a cooling experiment

Lab 01 diagnostic: build, inspect, and test a temperature model.

1 Observe

A warm sensor
cools toward room
temperature.



2 Model

Newton cooling model

$$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$$

Inputs

$$T_0 = 80\text{ °C}$$

$$T_{\text{env}} = 22\text{ °C}$$

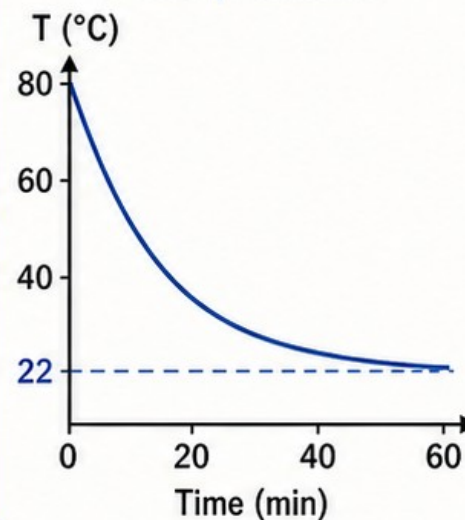
$$k = 0.08\text{ min}^{-1}$$

Time array

$$t_{\text{min}} = 0:2:60$$

3 Inspect

Model prediction



```
t_min = 0:2:60;
```

```
T_C = Tenv_C + (T0_C-Tenv_C).*exp(-k_per_min.*t_min);
```



Evidence checklist

- ☒ labelled plot
- ☒ first time within 1 °C
- ☒ initial value, bounds, trend



The code is short; the reasoning must remain visible.

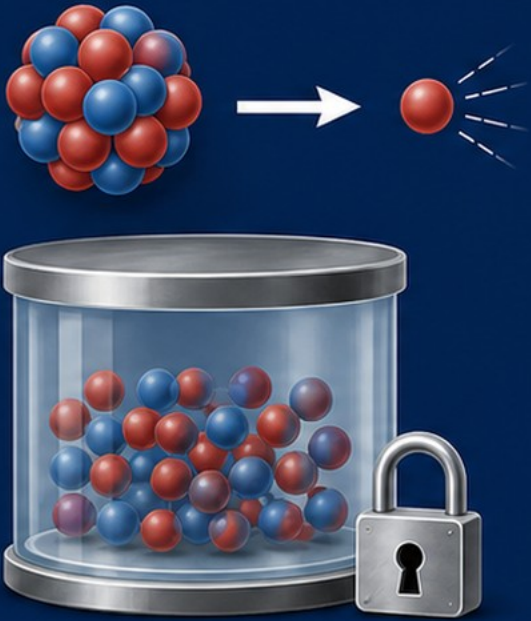


Diagnostic habits: arrays | element-wise operations | indexing | plotting | validation

Start with a prediction, then write the model

A closed sample with a constant half-life gives a controlled physical question.

Model



$$\frac{dN}{dt} = -\lambda N$$

$$N(0) = N_0$$



Question

A sample has half-life 6.0 h.
How many undecayed nuclei remain after 24 h?



Assumptions

- constant λ
- closed sample
- same decay probability per unit time



Predict

- After 6.0 h: 0.5 of N_0
- After 24 h: 0.0625 of N_0



A physical population should not become negative.

Map physics symbols to readable MATLAB names

A parameter map prevents notation and units from drifting during coding.

Physics	MATLAB name	meaning / unit
N_0	NO	initial population / nuclei
$T_{1/2}$	T_half_h	half-life / h
λ	lambda_per_h	decay constant / 1/h
t_{max}	tMax_h	final time / h
dt	dt_h	numerical step / h



Approved values

T_half_h = 6.0 h

tMax_h = 24 h

lambda_per_h = $\ln(2)/6.0 \approx 0.1155$ 1/h



Dimension check:

lambda_per_h $\times$ dt_h
is dimensionless.

The exact solution is a reference, not a shortcut

Use the benchmark to make approximation error visible.

$$N_{\text{exact}}(t) = N_0 \exp(-\lambda_{\text{per}_h} t)$$

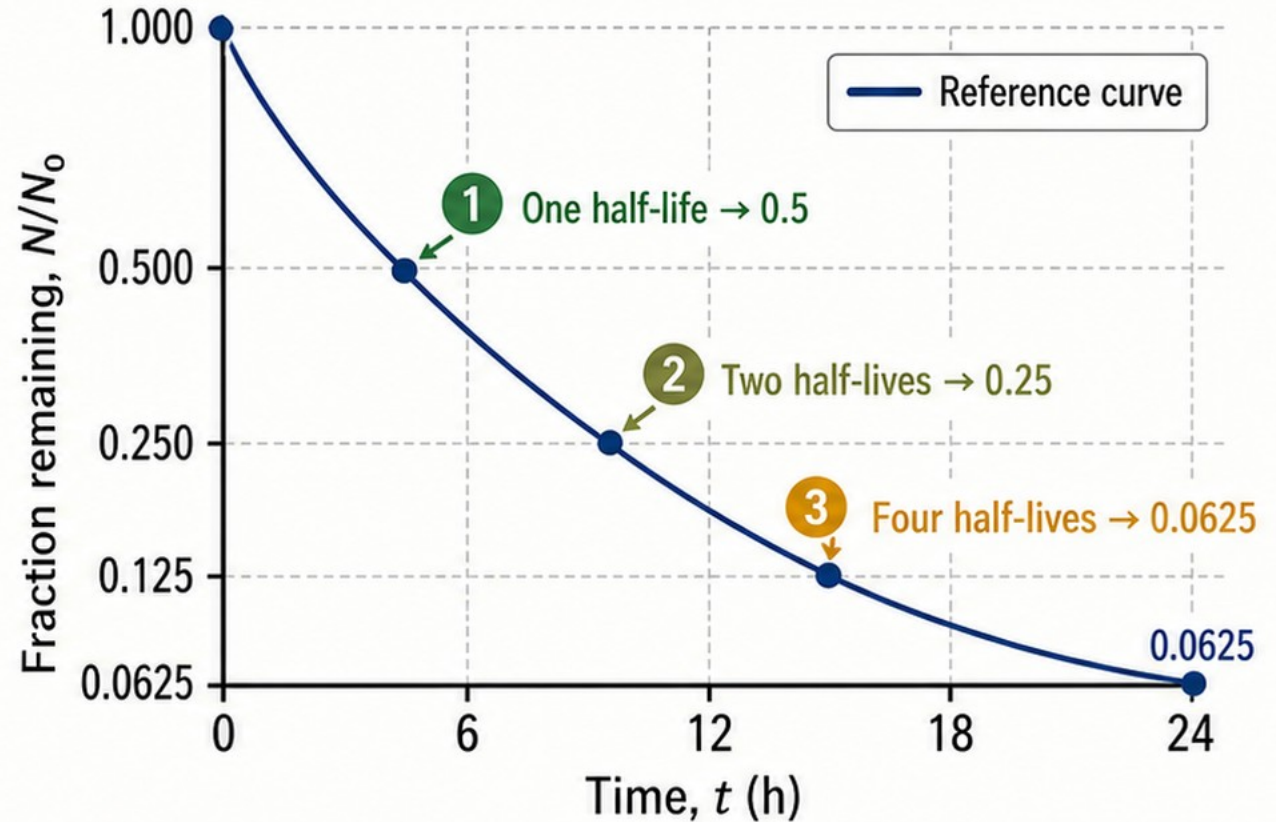
$$\lambda_{\text{per}_h} = \ln(2) / 6.0 \approx 0.1155 \text{ 1/h}$$

$$\text{At } t = 24 \text{ h: } N_{\text{exact}} / N_0 = 0.0625$$

1 One half-life $\rightarrow 0.5$

2 Two half-lives $\rightarrow 0.25$

3 Four half-lives $\rightarrow 0.0625$



Prediction before computing:
what should the numerical curve approach?



A benchmark does not remove the need for numerical work;
it gives the work something to answer to.

1 Build the time array before the loop

One time value must align with one stored population value.

```
1 t_h = 0:dt_h:tMax_h;  
2 N = zeros(size(t_h));  
3 N(1) = N0;
```

$$t_h(1) = 0$$

$$t_h(2) = dt_h$$

$$t_h(3) = 2 dt_h \dots$$

$$N(1) = N_0$$

N(2) after
one update

N(3) after
two updates

...



Preallocation

`zeros(size(t_h))` reserves the storage plan.



Initial condition

N(1) is N₀ because MATLAB indexing starts at 1.



Alignment

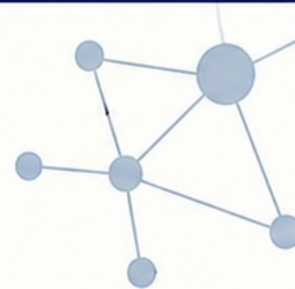
- $t_h(1) = 0 \leftrightarrow N(1) = N_0$
- $t_h(2) = dt_h \leftrightarrow N(2)$ after one update
- $t_h(3) = 2 dt_h \leftrightarrow N(3)$ after two updates



Common trap: N(0) is not a MATLAB index.

Turn the derivative into a next-value rule

Forward Euler keeps the model but approximates the time derivative.



1 Start with the model:

$$\frac{dN}{dt} = -\lambda N$$



2 Approximate the slope:

$$\frac{N_{n+1} - N_n}{dt} \approx -\lambda N_n$$



3 Rearrange:

$$N_{n+1} = N_n (1 - \lambda dt)$$



Approximation

This is not the exact physics; it is a discrete method for carrying the model forward.



Smaller dt

usually improves the approximation here



More steps

increase the computational cost



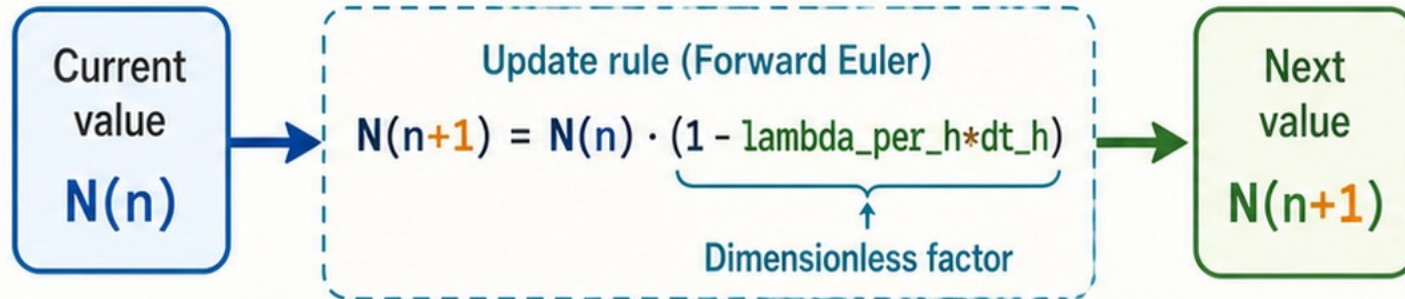
Warning: if $\lambda dt > 1$, the factor can become negative.



The loop is the algorithm written out

Read it as current value $\rightarrow$ update rule $\rightarrow$ next value.

```
for n = 1: numel(t_h)-1
    N(n+1) = N(n)*(1 - lambda_per_h*dt_h);
end
```



Checkpoint: what does one pass through the loop represent physically?

1

Read the current value $N(n)$

$N(n)$



current value

2

Multiply by the dimensionless factor
 $1 - \lambda_{per_h} \cdot dt_h$

$N(n)$

$\times$

$1 - \lambda_{per_h} \cdot dt_h$

3

Store the new value in $N(n+1)$

$N(n+1)$



next value



Common mistakes



Overwrite $N(n)$ instead of writing $N(n+1)$



Loop to $\text{numel}(t_h)$ and write past the end

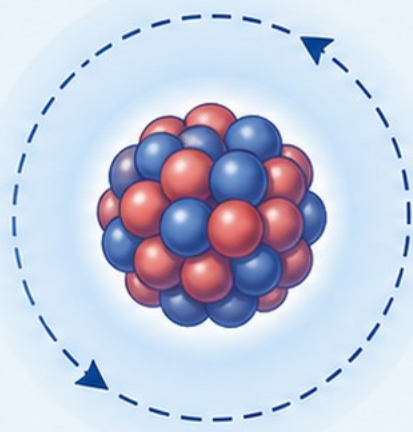


Forget $N(1) = N_0$ before the loop

Hold the physics fixed; vary one numerical choice

A fair timestep experiment changes resolution while keeping the model the same.

Physical model



$$N_0 = 1000$$

$$T_{\text{half}_h} = 6.0 \text{ h}$$

$$t_{\text{Max}_h} = 24 \text{ h}$$

$$\text{lambda_per_h} = \ln(2)/6.0$$

One variable
changed



All else
held fixed

Numerical resolution

dtValues_h = [2.0 1.0 0.5 0.1]

2.0
h

1.0
h

0.5
h

0.1
h

Same update rule; different number of steps

Step count over 0 to 24 h

12 steps

24 steps

48 steps

240 steps

dt = 2.0 h

dt = 1.0 h

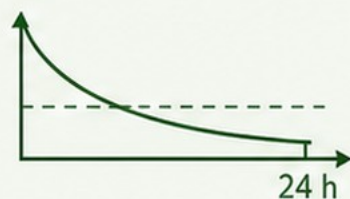
dt = 0.5 h

dt = 0.1 h

Record



Euler fraction at 24 h

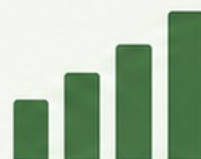


relative error against exact

$$\left| \frac{N_{\text{num}}(24 \text{ h}) - N_{\text{exact}}(24 \text{ h})}{N_{\text{exact}}(24 \text{ h})} \right| \times 100\%$$



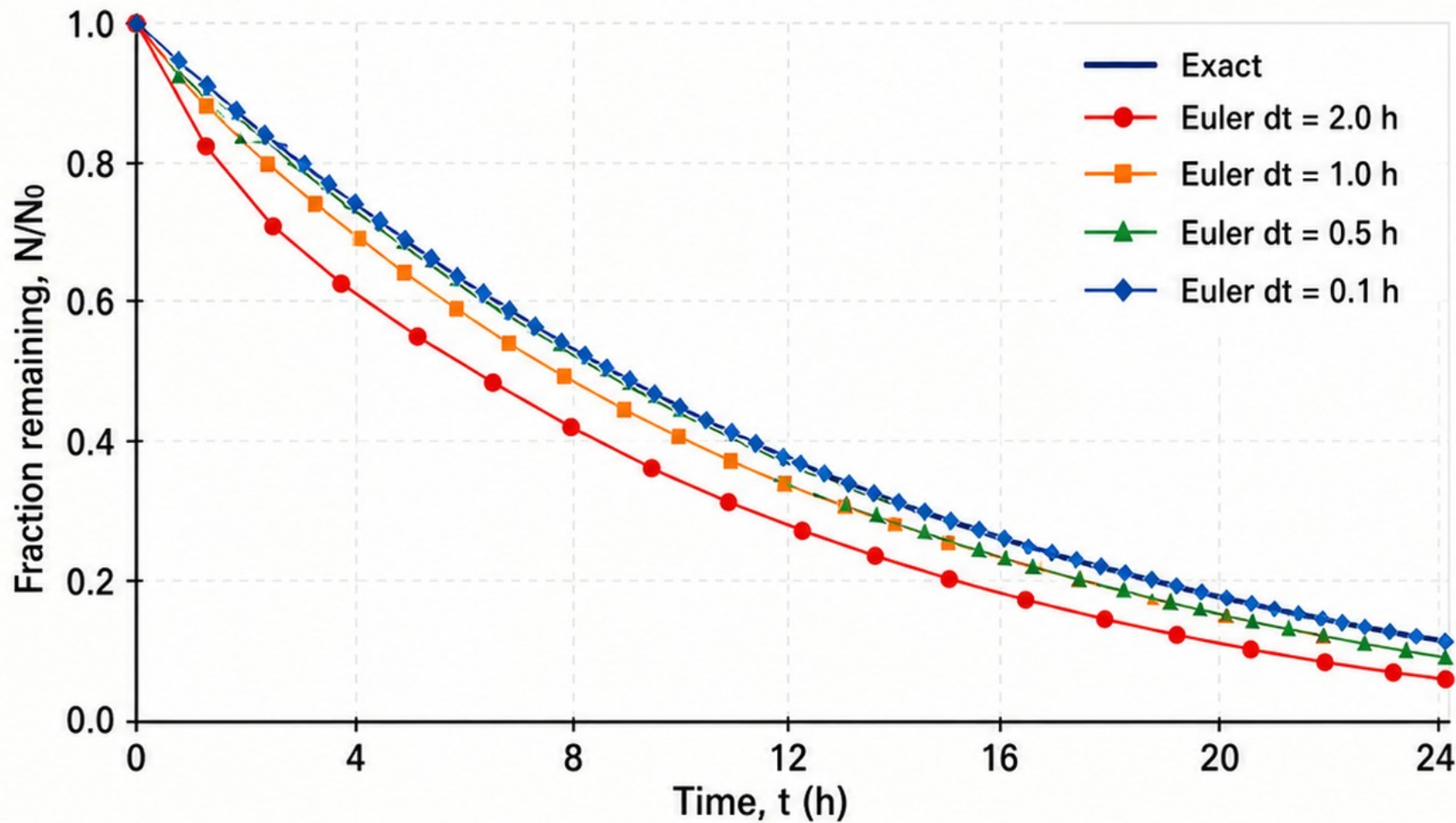
step count



Cost is part of
the result.

Coarse and fine Euler curves separate from the reference

The visual comparison shows why resolution is a modelling decision.



Read the shape

The visual comparison shows why resolution is a modelling decision.



Coarse

Coarse Euler steps fall below the exact curve at 24 h.



Fine

Fine steps track the reference more closely.

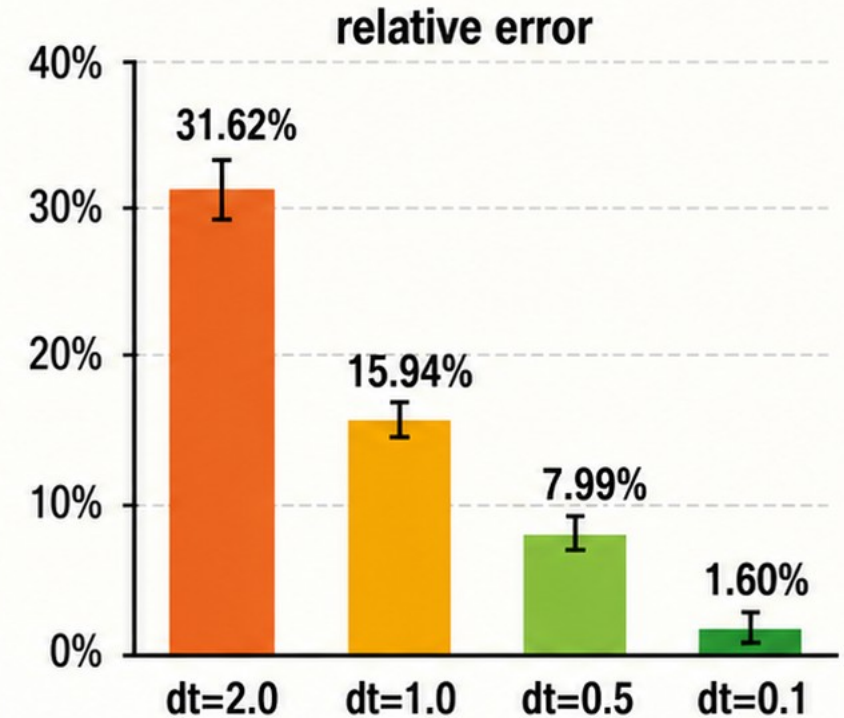


Question: which curve would you report first – and what evidence would you add?

Smaller dt improves the result — at a measurable cost

Approved checkpoints for $T_{\text{half}_h} = 6.0$ h and $t_{\text{Max}_h} = 24$ h

dt (h)	lambda*dt	Euler N/N0 at 24 h	relative error	steps
2.0	0.2310	0.042735	31.62%	12
1.0	0.1155	0.052536	15.94%	24
0.5	0.0578	0.057505	7.99%	48
0.1	0.0116	0.061499	1.60%	240



Exact reference at 24 h: $N/N_0 = 0.0625$

Trend: smaller dt → lower error, more steps

A numerical result includes accuracy and cost.

Trust comes from a trail of checks

Validation is evidence that the implementation and interpretation agree with a trusted check.

1



Physics

Question, variables, assumptions, expected trend

2



Units

lambda and t use the same time unit

3



Algorithm

Update rule, initial condition, index range

4



Numerics

At least two timesteps and a relative error

5



Communication

Labels, units, legend, and one interpretation

```
assert(abs(N(1)-N0) < eps(N0))
```

```
assert(abs(N_exact_fraction(end)-0.0625) < 1e-12)
```

```
assert(all(isfinite(N)))
```

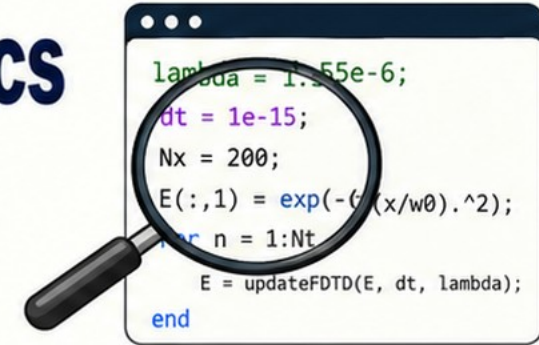
```
assert(all(N >= 0))
```



Warning: $\lambda \cdot dt > 1$ can make the Euler population negative.

AI can suggest syntax; it cannot certify physics

Responsible AI use means verification literacy, not blind acceptance.



A safe assistance pattern

- 1 Read the code line by line ✓
- 2 Check units and limiting cases ✓
- 3 Compare with N_{exact} and report error ✓
- 4 Rerun from a fresh MATLAB session ✓
- 5 State the test you performed ✓



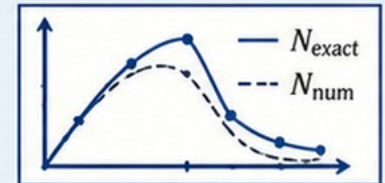
What the assistant cannot establish

- ⚠ Whether the physical assumptions fit the sample
- ⚠ Whether λ and dt use compatible units
- ⚠ Whether the chosen timestep is adequate for the question



Reproducibility record

parameters + units | $dt + t_{Max}$ | MATLAB version | validation evidence | labelled figure + caption



Checkpoint: what evidence makes your dt acceptable?



Leave with a model, a method, and a test you can rerun

The next task transfers the reasoning chain to a cooling model.

1 Takeaway 1



A computational result begins with a model and assumptions — not with a MATLAB command.

2 Takeaway 2



dt is part of the numerical method. Smaller is not automatically free.

3 Takeaway 3



Validation and interpretation are required parts of a simulation.



Next in the Week 1 package

- 1 Complete the individual entry check.
- 2 Open PHY4605_Week01_Onramp_Diagnostic.m.
- 3 Predict, compute, validate, and interpret the cooling model.



Exit ticket: model + assumption | algorithm | validation evidence



model



code



evidence